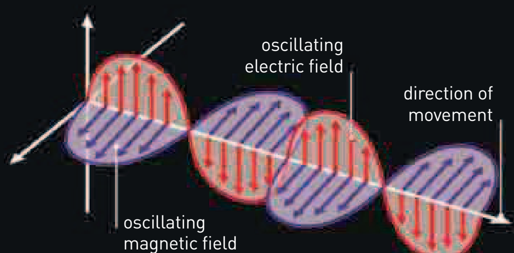
UNDERSTANDING ELECTROMAGNETIC RADIATION

DISCOVERIES IN THE 19TH CENTURY LED TO A NEW UNDERSTANDING OF THE NATURE OF RADIATION

Light, infrared and ultraviolet radiation, X-rays, gamma rays, microwaves, and radio waves all propagate through space at extremely high speed. They are all different forms of electromagnetic radiation, which can be understood as waves, but also as particles called photons.

By the 19th century, evidence suggested that light travels as waves and that wavelength determines the light waves' color. Two invisible forms of "light"—longer-wavelength infrared radiation (IR) and shorter-wavelength ultraviolet radiation (UV)—had also been discovered.



ELECTROMAGNETIC WAVE
In these self-propagating waves, oscillating electric and magnetic fields travel in the same direction but in perpendicular planes.

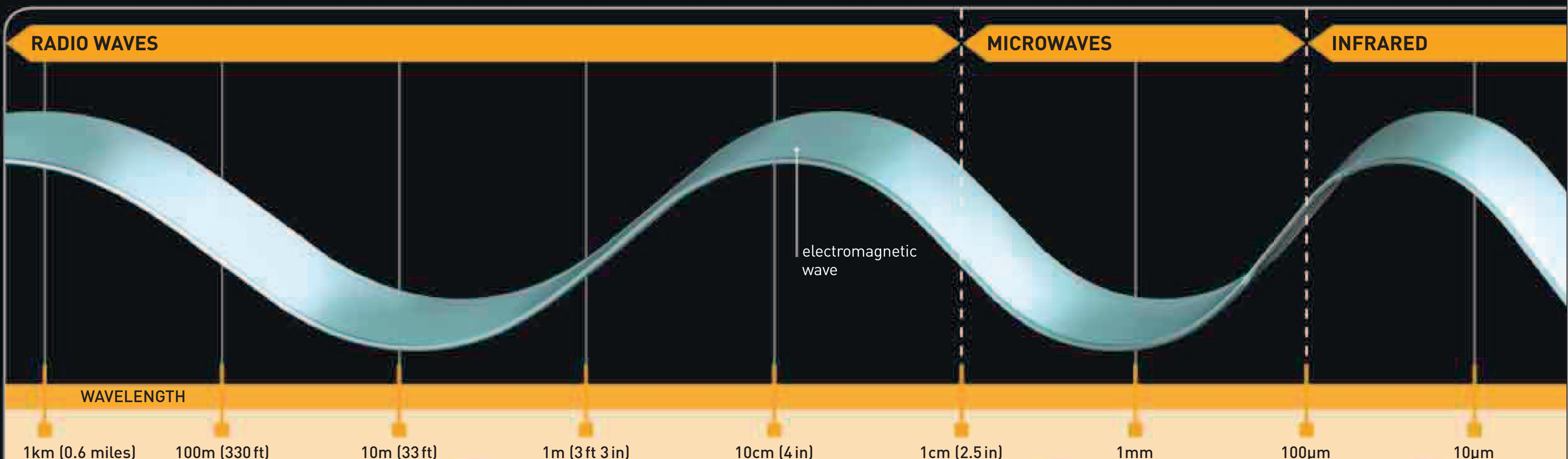
ELECTRIC AND MAGNETIC FIELDS

In the 1860s, British physicist James Clerk Maxwell formulated a set of equations that describes how electric fields produce magnetic fields and how magnetic fields produce electric fields. Maxwell realized that his formula was a "wave equation," describing wave motion. The speed of the waves described by the equation exactly matched the speed of light. Maxwell concluded that light is an "electromagnetic wave," and went on to predict the existence of other as yet unknown forms of radiation. Within 20 years, German physicist Heinrich Hertz had produced radio waves—electromagnetic waves with much longer wavelengths than light (see 1887).

JAMES CLERK MAXWELL
As well as theorizing the existence of electromagnetic waves, James Clerk Maxwell played a key role in interpreting the emerging science of thermodynamics, and took the first color photograph (see 1861).

2,050 miles

THE **DISTANCE** COVERED BY GUGLIEMO MARCONI'S FIRST **TRANSATLANTIC RADIO** SIGNAL IN 1901



Electromagnetic spectrum
Visible light represents a tiny part of the whole spectrum of electromagnetic radiation. Every part of the spectrum features in some way in the modern world; a few representative examples are shown here.

Radio telescope
Large dish telescopes that detect radio waves provide essential information about distant space.

Microwave oven
Produced by magnetrons, short-wavelength radio waves called microwaves are used to heat food.

Remote controller
Most remote controllers use coded infrared radiation to send instructions to electrical devices.